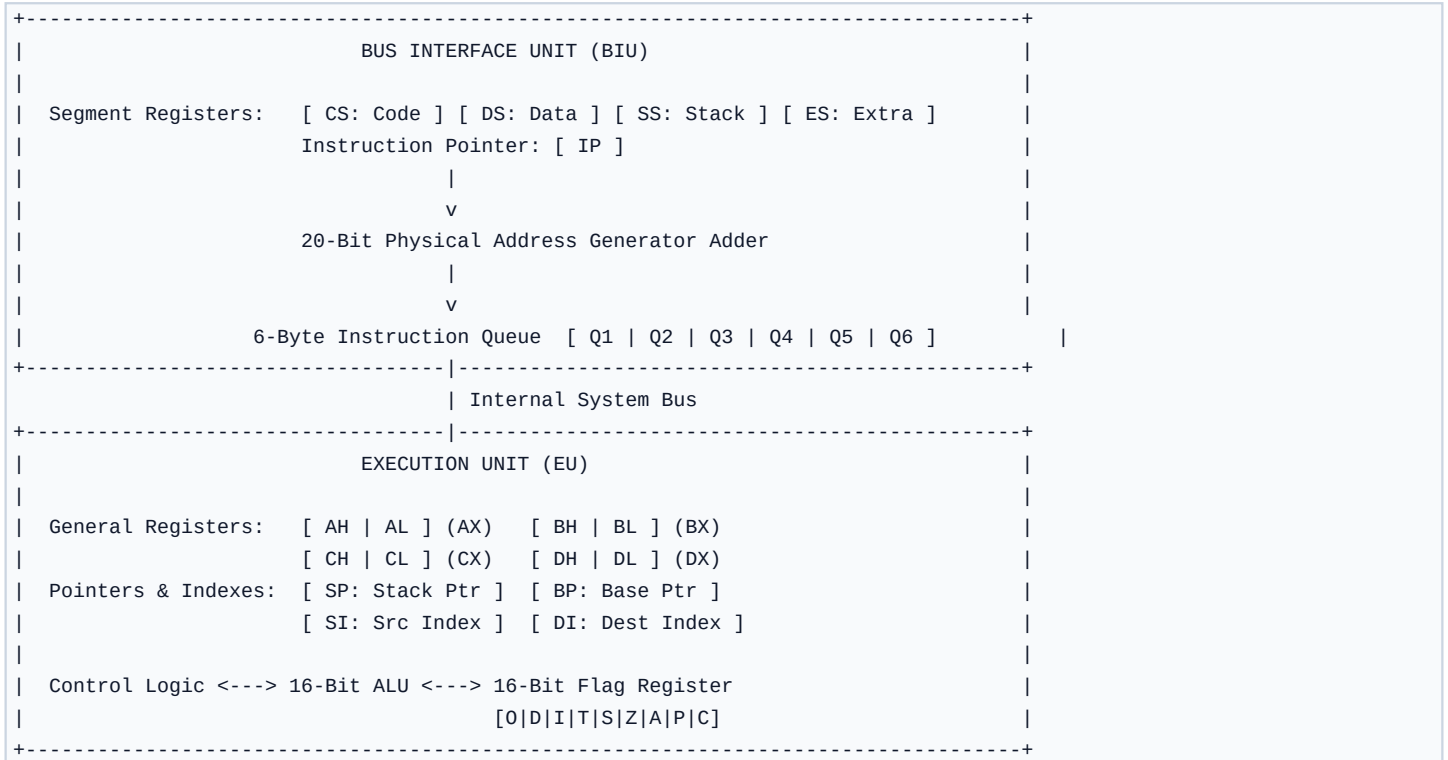


10-Mark Questions & Answers -- MPCA Unit 3: The Intel Microprocessor 8086 Architecture

Q1. Explain the internal architecture of 8086 in detail (BIU, EU, registers, flag register) with a neat labeled diagram.



1. Bus Interface Unit (BIU)

* Se

2. Execution Unit (EU)

* AL

Q2. Explain the functional pin diagram of 8086 microprocessor in detail, describing the function of each pin group (address/data bus, control signals, power supply).

	+---	---	+
GND	1	40	VCC (+5V)
AD14 <->	2	39	<-> AD15
AD13 <->	3	38	-> A16/S3
AD12 <->	4	37	-> A17/S4
AD11 <->	5	36	-> A18/S5
AD10 <->	6	35	-> A19/S6
AD9 <->	7	34	-> BHE' /S7
AD8 <->	8	33	-> MN/MX'
AD7 <->	9	32	-> RD'
AD6 <->	10	31	-> HOLD (RQ/GT0')
AD5 <->	11	30	-> HLDA (RQ/GT1')
AD4 <->	12	29	-> WR' (LOCK')

```

AD3 <-> | 13  28 | -> M/I0' (S2')
AD2 <-> | 14  27 | -> DT/R' (S1')
AD1 <-> | 15  26 | -> DEN' (S0')
AD0 <-> | 16  25 | -> ALE (QS0)
NMI --> | 17  24 | -> INTA' (QS1)
INTR --> | 18  23 | <- TEST'
CLK --> | 19  22 | <- READY
GND     | 20  21 | --> RESET
+-----+

```

1. Multiplexed Address/Data Bus (AD0 - AD15): Time-multiplexed lines carrying lower 16-bit memory address during T1 and 16-bit data during T2-T4.

2. Ac

Q3. Explain memory segmentation and banking in 8086 in detail with physical address calculation examples. Also explain why segmentation is used.

1. Why Memory Segmentation is Used

* All

2. Physical Address Calculation Formula & Example

* Example 1 (Instruction Fetch): CS = 3000H, IP = 0100H

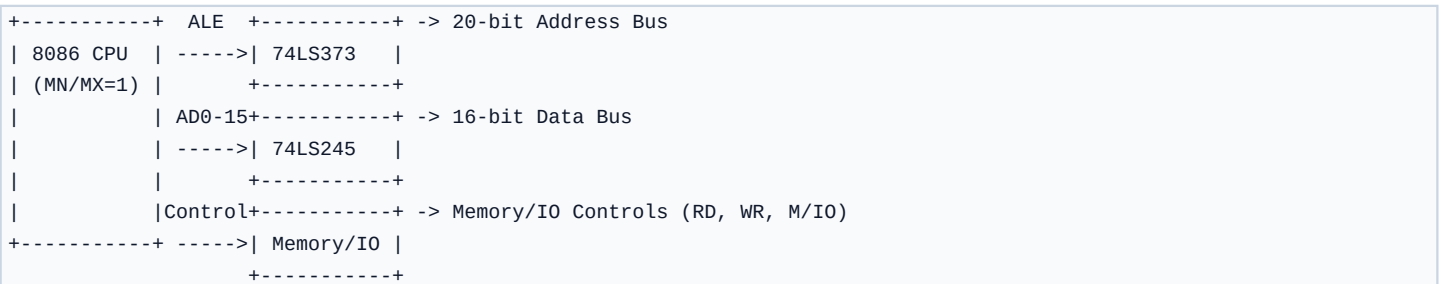
* Example 2 (Data Read): DS = 4500H, BX = 0250H

3. Memory Banking System (Even & Odd Banks)

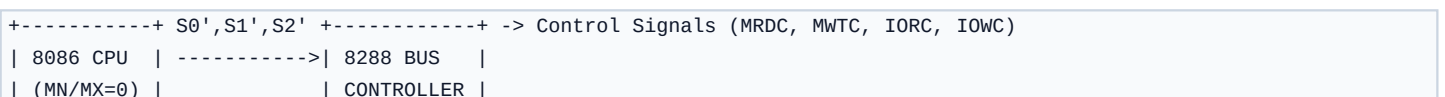
1MB PHYSICAL MEMORY SYSTEM		
/	EVEN BANK (512 KB)	ODD BANK (512 KB)
Address Line A0 = 0	Bus High Enable BHE' = 0	
Data Bus D0 - D7	Data Bus D8 - D15	

Q4. Explain the functioning of 8086 in both minimum mode and maximum mode with neat diagrams, and state the key differences.

1. Minimum Mode Block Diagram (MN/MX' = 1)



2. Maximum Mode Block Diagram (MN/MX' = 0)



```
+-----+ +-----+
```

Q5. Draw and explain in detail the timing diagrams for read and write operations in both minimum and maximum mode of 8086.

MINIMUM MODE READ TIMING DIAGRAM:

```
CLK      :  _-_-_-_-_-_-_-_-_-_-_-_-_-_-_-_-_-
           | T1 | T2 | T3 | T4 |
ALE      :  --_--_--_--_--_--_--_--_--_--_--_--
A16-A19  :  == Address == Status ==
AD0-AD15 :  == Address == High-Z == Data ==
RD       :  -_-_-_-_-_-_-_-_-_-_-_-_-_-_-_-_-
```

Q6. Explain the interrupt structure of 8086 in detail, including types of interrupts (hardware/software), interrupt vector table, and interrupt servicing procedure.

INTERRUPT VECTOR TABLE (00000H - 003FFH, 1KB RAM)

```
+-----+ 003FFH
| Vector 255 (Type 255) |
| ... |
| Vector 2 (NMI - Non-Maskable Pin) |
| Vector 1 (Single Step / Trap) |
| Vector 0 (Divide-by-Zero Error) |
+-----+ 00000H
```

Q7. Explain the process of demultiplexing the address/data bus in 8086 with a circuit diagram (using latch 74LS373).

